Ericsson PE Node Optical Transport Failure Resolution Procedure for Jayanagar Metro Site

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_5470

Description

The Optical Transport Failure domain on the Ericsson PE node, specifically at the Jayanagar Metro Site (Node ID: JAY-PE-09), can be attributed to a myriad of factors including but not limited to SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. This failure domain is particularly critical in the context of the PE role, as it directly impacts the network's ability to transport data across different domains. The underlying mechanics involve the interaction between the optical transport equipment and the fiber infrastructure, where any degradation or failure can lead to signal attenuation or complete loss, triggering alarms such as ALARM_5470. The Ericsson equipment, with its unique CLI dialect, provides a set of commands to diagnose and resolve these issues, which are pivotal in maintaining the health and reliability of the optical transport network. Understanding the intricacies of the optical transport layer, including the principles of wavelength division multiplexing (WDM) and the specifications of the SFP/XFP modules, is essential for effective troubleshooting and resolution. The adherence to the Ericsson CLI dialect is crucial for navigating the device's configuration and diagnostic capabilities, ensuring that any commands executed are compatible and effective in resolving the optical transport failure.

Pre-requisite Checks

- Dispatch remote hands to Jayanagar Metro Site for physical inspection
- Verify redundant link status to ensure no single point of failure
- Check environmental conditions such as temperature and humidity
- Confirm power supply to the affected node and its components

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to aging or environmental factors	High
Fiber cuts or damage leading to signal loss	Medium
Rx/Tx power loss due to equipment malfunction or configuration issues	Low

Diagnostic Commands & Expected Outputs

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Command: show optical-transport interface 1/1, Expected Output:
Interface status, including power levels and error counts,
Description: This command is used to isolate issues in the optical
transport interface, providing critical information on the current
state of the optical signals, which can indicate problems such as
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signal degradation or complete loss.

Command: show sfp detail, Expected Output: Detailed information about the SFP/XFP modules, including temperature, voltage, and current, Description: The detailed SFP/XFP module information helps in diagnosing hardware-related issues, such as overheating or power supply problems, which can lead to signal degradation or failure.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Module, Command: replace sfp 1/1, Description: Replacing the degraded SFP/XFP module with a new one can resolve issues related to signal degradation due to aging or environmental factors. This step involves physically removing the old module and inserting a new one, ensuring that the replacement module is compatible with the Ericsson equipment and the specific interface.

Step 2: Step: Adjust Rx/Tx Power Levels, Command: configure optical-transport interface 1/1 power-level -10, Description: Adjusting the Rx/Tx power levels can help in resolving issues related to signal strength. This step requires careful consideration of the optimal power levels for the specific optical transport configuration to avoid signal degradation or overload.

Post-Check Verification

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Command: show optical-transport interface 1/1, Expected Output: Updated interface status reflecting changes made during resolution, Description: Verifying the interface status after applying the resolution steps ensures that the optical transport failure has been successfully addressed and the network is operational within expected parameters.

Command: show alarm, Expected Output: List of current alarms, with ALARM_5470 cleared if resolution was successful, Description: Checking the alarm list provides confirmation that the specific alarm related to the optical transport failure has been cleared, indicating a successful resolution of the issue.
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Roll-back Procedures

- Step: Revert to Previous SFP/XFP Module, Command: replace sfp 1/1 with previous-module, Description: If the replacement SFP/XFP module does not resolve the issue or introduces new problems, reverting to the previous module can help in isolating if the issue was module-specific. This involves physically removing the new module and reinserting the original one.

- Step: Reset Rx/Tx Power Levels to Default, Command: configure optical-transport interface 1/1 power-level default, Description: If adjusting the Rx/Tx power levels does not resolve the issue or causes additional problems, resetting these levels to their default values can help in reverting to a known good state. This step requires knowledge of the default power level settings for the specific Ericsson equipment and optical transport configuration.